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Global Guidelines and Trends in HPV Vaccination for Cervical Cancer Prevention

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Cervical cancer remains a significant global health challenge, with persistent infection by oncogenic human papillomavirus (HPV) types identified as a necessary condition for its development. HPV vaccines have emerged as a crucial tool for preventing precancerous genital lesions and cervical cancer linked to specific HPV types. Over recent years, leading oncology and gynecology societies across the United States, Europe, and Australia have issued clinical guidelines aimed at optimizing the implementation of population-based HPV vaccination programs.

This narrative review synthesizes key recommendations for HPV vaccination programs, focusing on guidelines published between 2019 and 2024. A total of 16 guidelines were analyzed, revealing unanimous support for vaccinating children – ideally before the age of 15 years – to ensure protection prior to sexual activity. Most recommendations emphasize the vaccination of girls and boys, with the 9-valent Gardasil 9 vaccine identified as the preferred option. A major milestone was reached in 2022, when the World Health Organization proposed a single-dose regimen as an effective strategy, offering potential benefits in public health accessibility and program efficiency.

This article aims to review and compare current international guidelines for vaccination programs for human papilloma virus.

Keywords: **Cancer Vaccines • Papillomavirus Infections • Vaccines**

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Introduction

Cervical cancer is the fourth most common cancer among women worldwide and the fourth leading cause of cancer-related deaths in women [1]. In 2008, German scientist Harald zur Hausen was awarded the Nobel Prize for his discovery of human papillomaviruses (HPVs) as causative agents of cervical cancer [2]. Persistent infection with high-risk HPV types is now recognized as a necessary condition for the development of cervical cancer [3]. While HPV infection alone is not sufficient to cause cancer, vaccines targeting specific oncogenic HPV types have been developed and approved for the prevention of precancerous lesions of the genital tract and cervical cancer [4,5].

Two types of HPV vaccines are currently available for use: the 2-valent recombinant vaccine Cervarix, which protects against HPV types 16 and 18 [4], and the 9-valent recombinant Gardasil 9 vaccine, which provides broader protection against HPV types 6, 11, 16, 18, 31, 33, 45, 52, and 58 [5]. The latter has largely replaced the earlier 4-valent Gardasil/Silgard vaccine. These vaccines represent a crucial tool for reducing the burden of cervical cancer by preventing infections with oncogenic HPV types linked to its development.

Globally, cervical cancer remains a significant public health challenge, with over 600 000 new cases and 340 000 deaths reported annually [1]. Most cases occur in low- and middle-income countries, where limited access to screening and vaccination contributes to higher incidence and mortality rates. Key studies, such as those demonstrating the high efficacy of HPV vaccines in preventing cervical intraepithelial neoplasia (CIN) caused by high-risk HPV types, have provided a robust foundation for vaccine implementation programs [4,5].

International scientific societies and health organizations have issued clinical guidelines to guide the implementation of population-based HPV vaccination programs. These strategies often differ between countries, reflecting variations in the evidence base, healthcare infrastructure, and local epidemiology. However, all guidelines emphasize the critical role of vaccination in preventing HPV infections and cervical cancer.

The purpose of this article is to review, analyze, and summarize current clinical recommendations for implementing population-based HPV vaccination programs. By synthesizing guidelines from prominent oncology and gynecology societies in the United States, Europe, and Australia, this review aims to provide an updated perspective on strategies to prevent cervical cancer through effective vaccination programs.

Overview of HPV Vaccination Guidelines

In this analysis, we examined recommendations issued between 2019 and 2024, focusing on several key vaccination parameters. These included the optimal or priority age for initiating vaccination within the target population, the inclusivity of the vaccination target group in terms of sex, and the maximum age for catch-up vaccinations. Additionally, we considered the preferred vaccine formulations and the recommended dosing schedules. The review included guidelines published in English and Polish, and we intentionally excluded recommendations tailored to special populations, such as individuals with immunodeficiencies.

The search identified 16 recommendations from various scientific societies regarding HPV vaccination. These include guidelines from the Advisory Committee on Immunization Practices, the American Academy of Family Physicians, the American Cancer Society, and the American College of Obstetricians and Gynecologists. European organizations such as the Arbeitsgemeinschaft der Wissenschaftlichen Medizinischen Fachgesellschaften, European Society of Gynecologic Oncology, and European Federation for Colposcopy were also represented.

Additionally, recommendations of national authorities such as the Australian Government Department of Health, Center for Disease Prevention and Control, Haute Autorité de Santé, Health Council of the Netherlands, and Joint Committee on Vaccination and Immunization provided relevant guidance. Contributions from the National Cancer Institute and Polish societies, including the Polish Society for Colposcopy and Cervical Pathophysiology and other Polish scientific organizations, were also reviewed. Further recommendations came from the Royal College of Physicians of Ireland and the National Immunization Advisory Committee. Finally, the guidelines of the World Health Organization (WHO) offered global perspectives on HPV vaccination strategies.

Justification for Implementing Protective Vaccinations Against HPV

All found recommendations of scientific societies remain consistent with regard to the validity of vaccination against HPV. The main reason for the general recommendation of vaccinations against this virus is increased protection against infections and, consequently, reduced risk of cancer, mainly cervical cancer. A detailed list of the recommendations and their comparisons are presented in **Table 1**.

Age and Sex of Vaccination Target Population

Optimal/Priority Age of Vaccination Target Population

Five of the 16 guidelines indicate the need to implement vaccination as early as age 9 years, the lowest possible age allowed

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Table 1. List of recommendations included in this study.

Organization/ year	Optimal/priority age of vaccination target population	Sex of the target population for vaccination (F, female; M, male)	Upper age of target population for catch-up vaccination	Recommended vaccination	Administration scheme
Australian Government Department of Health, 2023 [6]	12-13 years	F and M	25 years	9vHPV	1 dose
Joint Committee on Vaccination and Immunization, 2023 [7]	12-13 years	F and M	25 years	9vHPV	2 doses
Polish Society for Colposcopy and Cervical Pathophysiology, 2023 [8]	12-13 years	F	26 years	9vHPV	1 or 2 doses
Royal College of Physicians of Ireland/National Immunization Advisory Committee, 2023 [9]	12-13 years	F and M	25 years	9vHPV	1 dose
Health Council of the Netherland, 2022 [10]	9 years	F and M	26 years	–	2 doses
Polish Scientific Societies, 2022 [11]	11-13 years	F	–	Based on an independent pharmacoeconomic analysis	–
World Health Organization, 2022 [12]	9-14 years	F	20 years	Based on significant local epidemiological data on HPV infections	1 or 2 doses
American Academy of Family Physicians, 2021 [13]	11-12 years	F and M	26 years	–	2 doses
Arbeitsgemeinschaft der Wissenschaftlichen Medizinischen Fachgesellschaften, 2021 [14]	9-14 years	F and M	26 years	–	2 doses
Center for Disease Prevention and Control, 2021 [15]	11-12 years	F and M	26 years	–	2 doses
National Cancer Institute, 2021[16]	11-12 years	F and M	26 years	–	2 doses
American College of Obstetricians and Gynecologists, 2020 [17]	11-12 years	F and M	26 years	–	–
American Cancer Society, 2020 [18]	9-12 years	F and M	26 years	–	–
Haute Autorité de Santé, 2020 [19]	11-14 years	F and M	19 years	9vHPV	–
Advisory Committee on Immunization Practices, 2019 [20]	11-12 years	F and M	26 years	–	–
European Society of Gynecologic Oncology/ European Federation for Colposcopy, 2019 [21]	9-13 years	F and M	25 years	–	2 doses

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by the Characteristics of Medicinal Products of available vaccine preparations [4,5]. Most, 7 of 16, recommendations recommend starting vaccination at age 11 years, and all 4 of the most recent 2023 recommendations point to age 12 years as the optimal age to start HPV vaccination programs. In contrast, all of the recommendations found indicate that vaccination programs should be conducted among children before the age of 15 years, to achieve HPV immunity before the onset of sexual activity.

Sex of Vaccination Target Population

Most recommendations indicate that girls and boys should be vaccinated. Only 3 recommendations recommend starting vaccination among girls, while covering boys depends on financial feasibility and the vaccination of a willing population of girls.

Upper Age of the Target Population for Catch-Up Vaccination

Most recommendations consistently state that HPV vaccination is worth pursuing until age 25-26 years. None of the recommendations recommend vaccination beyond age 26 years.

Vaccine Preparation and Vaccine Administration Schedule

Recommended Vaccine Preparation

Prior to 2023, only the French recommendation issued by the Haute Autorité de Santé in 2020 [19] recommended the implementation of vaccination with a specific 9-valent vaccine preparation. The rest of the recommendations did not indicate the superiority of any of the available preparations. In contrast, all recommendations issued in 2023 consistently recommended vaccination with Gardasil 9.

Recommended Vaccine Administration Schedule

By 2022, all organizations and scientific societies were recommending that vaccination be administered in a schedule in accordance with the summaries of product characteristics, namely, in the population of children under 15 years of age in a 2-dose schedule and in a 3-dose schedule among those 15 years of age and older and in children with immunodeficiencies. The WHO in 2022 was the first organization to indicate in its guidelines that it is possible to use only 1 dose of HPV vaccine, which is advisable from a public health perspective, as it provides a comparable and individual level of protection, while being easier to implement than a 2-dose regimen. In addition, the WHO has indicated that performing a single vaccination schedule can influence more girls to be reached more quickly, thereby leading to faster development of community immunity [12]. Following the WHO's recommendation, 3 of the 4 recommendations found in 2023 indicate the appropriateness of performing vaccination in an off-label regimen involving only 1 dose of vaccine.

Global Strategies for HPV Vaccination

In 2020, the WHO published "Global strategy to accelerate the elimination of cervical cancer as a public health problem" [22], setting the goal of eliminating cervical cancer as a population problem by the end of the current century. Among other things, the first step of this vaccination strategy is to have 90% of girls fully vaccinated against HPV by the age of 15 by 2030. The European Cancer Organization [23] also developed the European Cancer Plan in 2020, encouraging all EU member states to implement gender-neutral vaccination programs. According to the European Cancer Organization's recommendations, HPV vaccination programs should be in place in all European countries by 2030, with a target vaccination rate of at least 90% of adolescents of both sexes.

Comparative Efficacy of HPV Vaccines

The European Centre for Disease Prevention and Control, in its 2020 guidelines [24], provided an indirect comparison of the efficacy of available HPV vaccine formulations. In the comparison of 9vHPV and 4vHPV formulations, conclusions from the analysis of immunogenicity studies indicate (1) equivalence of the 9vHPV vaccine and the 4vHPV variant with respect to HPV serotypes 6, 11, 16, and 18; (2) stronger immune response against additional serotypes 31, 33, 45, 52, and 58 contained in the 9vHPV vaccine, compared with that in the 4vHPV variant; and (3) stronger immunogenicity of the 9vHPV vaccine against vaccine serotypes in males and females aged 9-15 years than in females aged 16-26 years.

On the other hand, in the comparison of 4vHPV and 2vHPV vaccines, the European Centre for Disease Prevention and Control indicates (1) equivalent efficacy of 4vHPV and 2vHPV vaccines in males and females; (2) higher immunogenicity for the 4vHPV and 2vHPV formulations in males aged 9-15 years than in females aged 16-26 years, for the specific HPV types contained in each vaccine.

Effectiveness in Preventing Cervical Neoplasia

In the recently published network meta-analysis by Lin et al in 2023 [25], the authors compared the effectiveness of p/HPV vaccination in preventing the occurrence of cervical neoplasia grade 2 and worse (CIN 2+) associated with HPV infections. According to the results of this publication, vaccination with 2vHPV, compared with placebo, statistically significantly reduces the risk of CIN2+ lesions associated with HPV16 infection by 94% (RR=0.06, 95% CI: 0.02; 0.19) and those associated with HPV18 infection by 92% (RR=0.08, 95% CI: 0.01; 0.67). On the other hand, vaccination with 9vHPV, compared with placebo, has

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a statistically significant effect on reducing the risk of CIN2+ lesions associated with HPV16 infection by 99% (RR=0.0195% CI: 0.00; 0.80). With regard to lesions caused by the HPV18 strain, the result did not reach statistical significance.

Future Directions

Future efforts should prioritize expanding HPV vaccination coverage, particularly in underserved populations, to ensure equitable access. Research should focus on optimizing vaccination schedules, including validating single-dose regimens, and improving vaccine formulations to provide broader protection against HPV strains. Robust monitoring and evaluation systems are essential for assessing the long-term impact of vaccination programs and guiding necessary adjustments. Integrating HPV vaccination into broader health initiatives, such as school-based programs and cancer prevention strategies, can enhance acceptance and efficiency. Global collaboration and public education campaigns will also play a vital role in addressing disparities, reducing vaccine hesitancy, and accelerating progress toward the elimination of cervical cancer as a public health problem.

Conclusions

The recommendations from all analyzed scientific societies unanimously emphasize the importance of administering HPV vaccination to children before the onset of sexual activity. Over time, as new evidence has emerged demonstrating the effectiveness of HPV vaccination in preventing infections linked to cervical cancer, these guidelines have evolved. Recent recommendations increasingly advocate for the use of the 9-valent vaccine, which offers broader protection against a wider range of HPV variants.

A significant innovation in the latest guidelines is the endorsement of a single-dose vaccination regimen for population-based programs. This approach aims to simplify program logistics, making large-scale vaccination efforts more feasible. Moreover, promoting HPV vaccination as a highly protective measure achievable through just 1 dose has the potential to address a persistent challenge in population-based immunization programs: low participation rates and suboptimal vaccine coverage. By reducing logistical and accessibility barriers, this strategy could significantly enhance the effectiveness of HPV vaccination campaigns.

Institution Where Work Was Done

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Institutional Review Board Statement

Decision of Medical University of Warsaw (no. AKBE 240/2023).

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